

PROJECT

6.3

**CONNECT A WEB APP
WITH AURORA
DATABASE SERIES**

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Objective

Learn how to build and host a web application on EC2 that connects to an Amazon Aurora MySQL database.

Understand how real world applications store and retrieve user data through secure backend connections.

Services

Amazon Aurora (MySQL-Compatible) - A fully managed database engine used to store dynamic data from your web app.

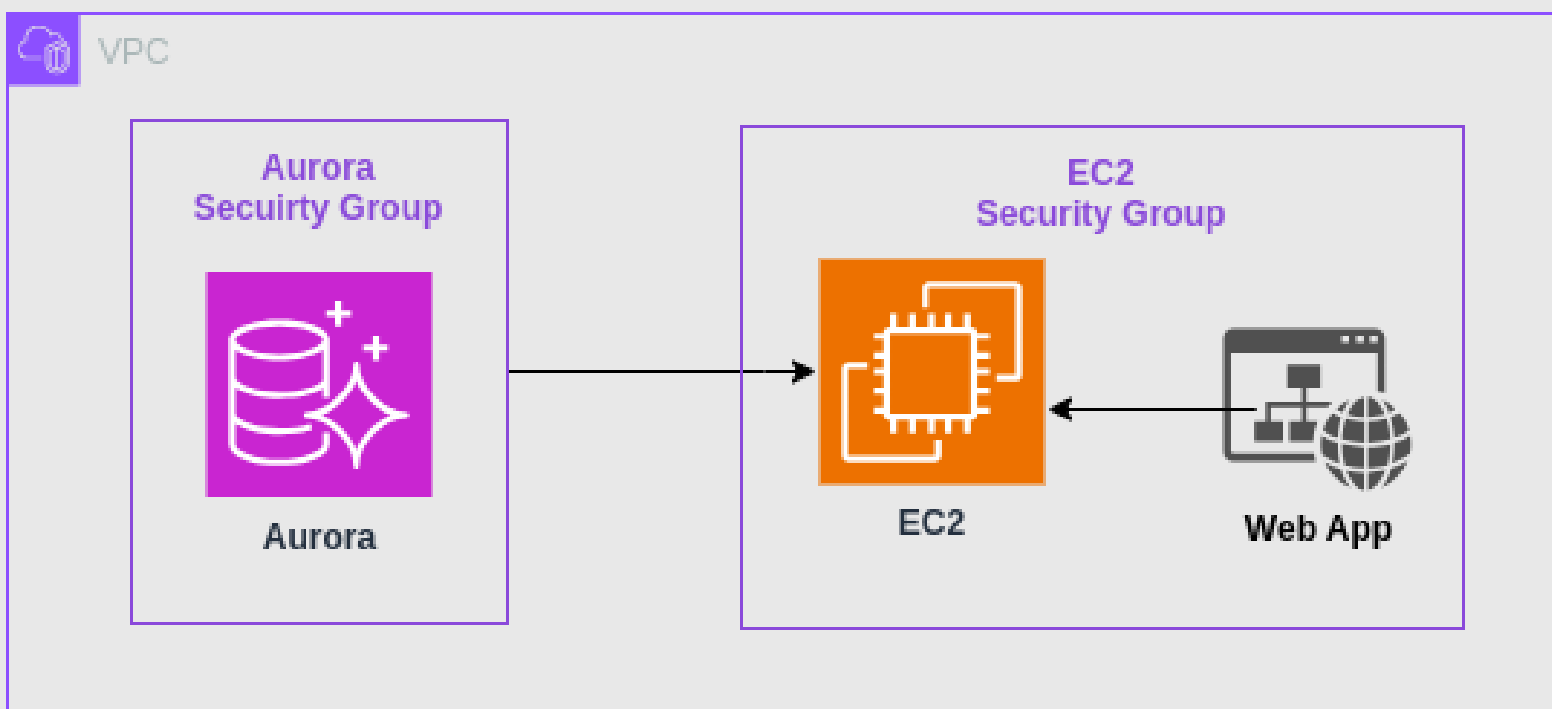
Amazon EC2 - Hosts your web app securely in the cloud, enabling interaction with users and backend services.

IAM (Identity and Access Management) - Manages user access and securely assigns roles and permissions.

Amazon VPC - Provides a private cloud network where your EC2 and Aurora services can securely communicate.

Security Groups - Act as digital firewalls that control traffic between the web app (EC2) and the database (Aurora).

Architecture Diagram



The web app runs on an EC2 instance inside a VPC, and the Aurora MySQL database is also placed in the same VPC. Security groups allow the EC2 instance to connect securely to Aurora over port 3306. This setup enables the web app to send user data directly to the database, all within a private, well protected network, just like in a real world cloud powered application.

Steps to Implement 1

1. Login with IAM User:

Log in to your AWS account using an IAM user with permissions for EC2 and RDS. This keeps your environment secure and avoids using the root user. It's your first step to safely accessing the AWS cloud.

2. Launch an EC2 Instance for Web App:

Head to the EC2 dashboard and launch a new instance within your VPC. Choose an Amazon Linux or Ubuntu AMI, attach a key pair, and allow HTTP, SSH access. This is where your web app will live and run.

3. Create an Aurora MySQL Database:

Go to the RDS console and create an Aurora MySQL DB in the same VPC. Keep it in a private subnet and make sure the security group allows inbound access on port 3306 from EC2. This acts as your app's secure data vault.

4. Build Your Web Application:

Inside your EC2, write a simple web app using PHP, Python, or Node.js. Link it to Aurora using credentials and the DB endpoint. This app will collect user input and store it in your database.

Steps to Implement 2

5. Make Your Web App Cool:

Add styling with HTML/CSS and improve how it looks and feels. Add input checks and success messages for a smooth experience. A polished frontend makes your app look real-world ready.

6. Test the Web App Connection:

Visit your EC2 public IP to open the app and try submitting some data. Check the Aurora database using the MySQL client to confirm it's stored. This proves your web app and database talk successfully.

7. Secure and Monitor:

Update your security groups to only allow trusted access and monitor logs. Use CloudWatch for tracking EC2 and Aurora activity. Regular reviews help you stay protected and performance-aware.

8. Clean Up Resources:

When done, terminate the EC2 instance and delete the Aurora cluster to avoid charges. Also remove unused IAM roles, security groups, and key pairs. This keeps your cloud setup neat and budget friendly.

Story Time

VPC (Your Cloud Neighborhood):

Think of the VPC as your cloud neighborhood where only invited friends can live and chat. Inside, both the web app (EC2) and Aurora database get cozy homes in private streets. This safe zone ensures no stranger sneaks in without a magic pass!

Aurora (The Magical Memory Vault):

Aurora is your enchanted vault that remembers everything from passwords to pizza orders. It's fast, smart, and only opens its golden gates to trusted guests. Without proper credentials, not even a whisper gets in!

EC2 (The Front-Door Hero):

Meet EC2 your web app's home and hero. It takes messages from users and walks them over to Aurora like a loyal messenger. But it can only enter the vault if its secret handshake (security group rule) matches perfectly.

IAM (The Access Wizard):

IAM is your trusted wizard who grants special access scrolls to heroes like EC2. Without IAM's magical permissions, no one can even enter the cloud gates. It's all about knowing who can do what, and where!

Security Groups (The Talking Rules):

Security groups are like the official rules posted on every castle door. They say who's allowed to talk to whom, and on which channels. When EC2 and Aurora both follow the rules, they become instant besties and share secrets all day long!

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THANK YOU!

